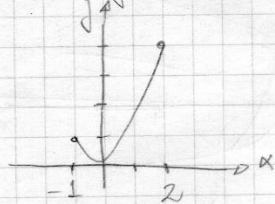


EJERCICIOS RESUELTOS – TP8: CURVAS, INTEGRALES DE LÍNEA, FUNCIÓN POTENCIAL

01)

a) $y = x^2$ entre $(-1, 1)$ y $(2, 4)$



$$\vec{g}_1: [-1, 2] \rightarrow \mathbb{R}^2 / \vec{g}_1(t) = (t, t^2) \quad \checkmark \text{ sentido}$$

$$\vec{g}_2: [-2, 1] \rightarrow \mathbb{R}^2 / \vec{g}_2(t) = (-t, t^2) \quad \checkmark \text{ sentido}$$

(con $u = -t$: $-1 \leq t \leq 2 \Rightarrow 1 \geq -t \geq -2$)

$$\begin{aligned} 1) a) L_1 &= \int_{-1}^2 \|\vec{g}_1'(t)\| dt = \int_{-1}^2 \|(1, 2t)\| dt = \int_{-1}^2 \sqrt{1+4t^2} dt = \\ &= \int_{-1}^2 2\sqrt{\frac{1}{4} + t^2} dt = 2 \left[\frac{1}{2} \left(t \sqrt{\frac{1}{4} + t^2} + \frac{1}{4} \cdot \right. \right. \end{aligned}$$

$$\begin{aligned} &\left. \operatorname{arsh} 2t \right) \Big|_{-1}^2 = 2 \sqrt{\frac{1}{4} + 4} + \frac{1}{4} \operatorname{arsh} 4 - (-1) \sqrt{\frac{1}{4} + 1} - \\ &- \frac{1}{4} \operatorname{arsh} (-2) = 2 \frac{\sqrt{17}}{2} + \frac{1}{4} \operatorname{arsh} 4 + \frac{\sqrt{5}}{2} - \frac{1}{4} \operatorname{arsh} (-2) = \\ &= \sqrt{17} + \frac{\sqrt{5}}{2} + \frac{\operatorname{arsh} 4}{4} + \frac{\operatorname{arsh} (2)}{4} \end{aligned}$$

$$= 2 \left(\frac{\sqrt{17}}{2} + \frac{1}{4} \operatorname{arsh} 4 + \frac{\sqrt{5}}{2} - \frac{1}{4} \operatorname{arsh} (-2) \right)$$

$$L_2 = \int_{-2}^1 \|\vec{g}_2'(t)\| dt = \int_{-2}^1 \sqrt{1+4t^2} dt =$$

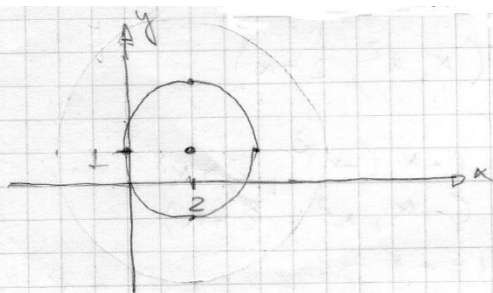
$$= 2 \left[\frac{1}{2} \left(t \sqrt{\frac{1}{4} + t^2} + \frac{1}{4} \operatorname{arsh} 2t \right) \Big|_{-2}^1 \right] =$$

$$= \sqrt{\frac{1}{4} + 1} + \frac{1}{4} \operatorname{arsh} 2 - (-2) \sqrt{\frac{1}{4} + 4} - \frac{1}{4} \operatorname{arsh} (-4) =$$

$$= \frac{\sqrt{5}}{2} + \frac{\operatorname{arsh} 2}{4} + \frac{\sqrt{17}}{2} + \frac{\operatorname{arsh} 4}{4} = \sqrt{17} + \frac{\sqrt{5}}{2} +$$

$$+ \frac{\operatorname{arsh} 4}{4} + \frac{\operatorname{arsh} 2}{4}$$

1b)



$$(x-2)^2 + (y-1)^2 = 2^2$$

$$\left(\frac{x-2}{2}\right)^2 + \left(\frac{y-1}{2}\right)^2 = 1$$

$$\begin{cases} \frac{x-2}{2} = \cos t & \Rightarrow x = 2\cos t + 2 \\ \frac{y-1}{2} = \sin t & \Rightarrow y = 2\sin t + 1 \end{cases}$$

(cos u = -t)

$$\bar{g}_1: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_1(t) = (2\cos t + 2, 2\sin t + 1) \quad \uparrow$$

$$\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_2(t) = [2\cos(-t) + 2, 2\sin(-t) + 1]$$

Other forms: $\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_2(t) = [2\sin t + 2, 2\cos t + 1]$

$$1b) L_1 = \int_0^{2\pi} \|(-2\sin t, 2\cos t)\| dt = \int_0^{2\pi} \sqrt{4\sin^2 t + 4\cos^2 t} dt =$$

$$= \int_0^{2\pi} 2 dt = 2(2\pi) = 4\pi$$

$$L_2 = \int_0^{2\pi} \|(2\sin(-t), -2\cos(-t))\| dt = \int_0^{2\pi} \sqrt{4\sin^2(-t) + 4\cos^2(-t)} dt$$

$$= \int_0^{2\pi} 2 dt = 4\pi$$

$$c) \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1 \Rightarrow \frac{x}{a} = \cos t$$

$$\frac{y}{b} = \sin t$$

$$\bar{g}_1: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_1(t) = (a \cos t, b \sin t)$$

$$\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_2(t) = [a \cos(-t), b \sin(-t)]$$

Other forms: $\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^2 / \bar{g}_2(t) = (a \sin t, b \cos t)$

$$1)c) L_1 = \int_0^{2\pi} \|(-a \sin t, b \cos t)\| dt = \int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} dt$$

$$L_2 = \int_0^{2\pi} \|(a \sin(-t), -b \cos(-t))\| dt =$$

$$= \int_0^{2\pi} \sqrt{a^2 \sin^2(-t) + b^2 \cos^2(-t)} dt =$$

$$= \int_0^{2\pi} \sqrt{a^2 \underbrace{\sin(-t) \sin(-t)}_{-\sin t} + b^2 \underbrace{\cos(-t) \cos(-t)}_{\cos t}} dt =$$

$$= \int_0^{2\pi} \sqrt{a^2 \sin^2 t + b^2 \cos^2 t} dt$$

$$d) \overline{AB} = (3, 2, 1) - (2, 3, -1) = (1, -1, 2)$$

$$(x, y, z) = (2, 3, -1) + t(1, -1, 2) \quad t \in \mathbb{R}$$

$$\vec{g}_1: [0, 1] \rightarrow \mathbb{R}^3 / \vec{g}_1(t) = (2+t, 3-t, -1+2t)$$

$$\vec{g}_2: [-1, 0] \rightarrow \mathbb{R}^3 / \vec{g}_2(t) = (2-t, 3+t, -1-2t)$$

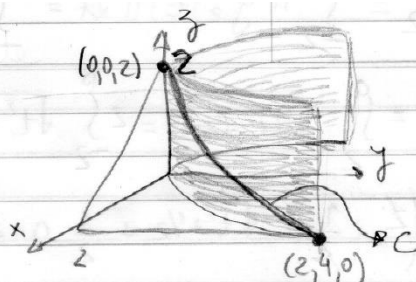
$$(\text{con } u = -t : 0 \leq t \leq 1 \Rightarrow 0 \geq -t \geq -1)$$

$$1)d) L_1 = \int_0^1 \|(1, -1, 2)\| dt = \int_0^1 \sqrt{1+1+4} dt = \sqrt{6}$$

$$L_2 = \int_{-1}^0 \|(-1, 1, -2)\| dt = \int_{-1}^0 \sqrt{1+1+4} dt = \sqrt{6} (0 - (-1)) = \sqrt{6}$$

$$1) e) \begin{cases} y = x^2 \\ x+z=2 \end{cases} \text{ 1° octante}$$

$$\begin{cases} y = x^2 \\ z = 2-x \end{cases} \text{ 1° oct}$$



$$\bar{g}_1(t) = (t, t^2, 2-t)$$

$$1^\circ \text{ oct} : \begin{cases} t \geq 0 \rightarrow t \geq 0 \\ t^2 \geq 0 \text{ vale } \forall t \in \mathbb{R} \\ 2-t \geq 0 \rightarrow 2 \geq t \end{cases} \Rightarrow 0 \leq t \leq 2$$

$$\bar{g}_1(t) : [0, 2] \rightarrow \mathbb{R}^3 / \bar{g}_1(t) = (t, t^2, 2-t)$$

$$\text{Pto inicial: } \bar{g}_1(0) = (0, 0, 2)$$

$$\text{Pto final: } \bar{g}_1(2) = (2, 4, 0)$$

$$\text{Consideramos } u = -t \rightarrow \bar{g}_2 = (\bar{g}_1 \circ u)$$

$$0 \leq t \leq 2 \Rightarrow 0 \geq \underbrace{-t}_u \geq -2$$

$$\bar{g}_2 : [-2, 0] \rightarrow \mathbb{R}^3 / \bar{g}_2(t) = (-t, (-t)^2, 2+t) = (-t, t^2, 2+t)$$

$$\text{Pto inicial: } \bar{g}_2(-2) = (2, 4, 0)$$

$$\text{Pto final: } \bar{g}_2(0) = (0, 0, 2)$$

Cálculo longitudinal de la curva

$$L_1 = \int_0^2 \|\bar{g}'_1(t)\| dt = \int_0^2 \|(1, 2t, -1)\| dt = \int_0^2 \sqrt{1+4t^2+1} dt =$$

$$= \int_0^2 \sqrt{4t^2+2} dt = 2 \int_0^2 \sqrt{t^2+1/2} dt =$$

$$= \left[\frac{1}{2} \left(t \sqrt{t^2+1/2} + \frac{1}{2} \operatorname{arsh} \sqrt{2} t \right) \right]_0^2 =$$

$$= 3\sqrt{2} + \operatorname{arsh}(2\sqrt{2}) / 2$$

$$L_2 = \int_{-2}^0 \|\bar{g}'_2(t)\| dt = \int_{-2}^0 \|(-1, 2t, 1)\| dt =$$

$$= \int_{-2}^0 \sqrt{4t^2+2} dt = 2 \int_{-2}^0 \sqrt{t^2+1/2} dt =$$

$$= \left[\frac{1}{2} \left(t \sqrt{t^2+1/2} + \frac{1}{2} \operatorname{arsh} \sqrt{2} t \right) \right]_{-2}^0 =$$

$$= 3\sqrt{2} + \operatorname{arsh}(2\sqrt{2}) / 2$$

$$f) \begin{cases} x^2 + y^2 + z^2 = 8 \\ z = \sqrt{x^2 + y^2} \end{cases}$$

$$x^2 + y^2 + x^2 + y^2 = 8 \Rightarrow 2x^2 + 2y^2 = 8 \Rightarrow x^2 + y^2 = 4$$

$$x = 2 \cos t$$

$$y = 2 \sin t$$

$$z = \sqrt{4 \cos^2 t + 4 \sin^2 t} = \sqrt{4} = 2$$

$$\bar{g}_1: [0, 2\pi] \rightarrow \mathbb{R}^3 / \bar{g}_1(t) = (2 \cos t, 2 \sin t, 2)$$

$$\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^3 / \bar{g}_2(t) = (2 \cos(-t), 2 \sin(-t), 2)$$

the param: $\bar{g}_2: [0, 2\pi] \rightarrow \mathbb{R}^3 / \bar{g}_2(t) = (2 \sin t, 2 \cos t, 2)$

$$f) L_1 = \int_0^{2\pi} \|(-2 \sin t, 2 \cos t, 0)\| dt = \int_0^{2\pi} \sqrt{4 \sin^2 t + 4 \cos^2 t} dt =$$

$$= 2 \int_0^{2\pi} dt = 4\pi$$

$$L_2 = \int_0^{2\pi} \|(2 \sin(-t), -2 \cos(-t), 0)\| dt =$$

$$= \int_0^{2\pi} \sqrt{4 \sin^2(-t) + 4 \cos^2(-t)} dt = \int_0^{2\pi} 2 dt = 4\pi$$

$$g) \quad u=3 \rightarrow \bar{X}_3 = (9r, 3-r, r+3)$$

$$(9r, 3-r, r+3) = (9, 2, 4) \Rightarrow r=1$$

$$(9r, 3-r, r+3) = (18, 1, 5) \Rightarrow r=2$$

$$\bar{g}_1: [1, 2] \rightarrow \mathbb{R}^3 / \bar{g}_1(r) = (9r, 3-r, r+3)$$

$$\bar{g}_2: [-2, -1] \rightarrow \mathbb{R}^3 / \bar{g}_2(r) = (-9r, 3+r, -r+3)$$

(con $u = -r$: $1 \leq u \leq 2 \Rightarrow -1 \geq -u \geq -2$)

$$9) L_1 = \int_1^2 \|(9, -1, 1)\| \, dr = \int_1^2 \sqrt{81+1+1} \, dr = \sqrt{83}$$

$$L_2 = \int_{-2}^{-1} \|(-9, 1, -1)\| \, dr = \int_{-2}^{-1} \sqrt{81+1+1} \, dr = \sqrt{83} (-1+2) = -\sqrt{83}$$

2)

$y=x$
 $x^2+y^2=4 \Rightarrow$
 $2x^2=4$
 $x^2=2$
 $|x|=\sqrt{2}$

$B = (0, 2)$
 $C = (0, 0)$
 $A = (\sqrt{2}, \sqrt{2})$

$\frac{\pi}{4} \leq t \leq \frac{\pi}{2}$
 $\widehat{AB}: \vec{q}(t) = (2\cos t, 2\sin t)$
 $\widehat{BC}: h(t) = (0, t) \quad 0 \leq t \leq 2$
 $\widehat{CA}: m(t) = (t, t) \quad 0 \leq t \leq \sqrt{2}$

$$L = L_{\widehat{AB}} + L_{\widehat{BC}} + L_{\widehat{CA}} =$$

$$= \int_{\pi/4}^{\pi/2} \|\vec{q}'(t)\| \, dt + \int_0^2 \|h'(t)\| \, dt + \int_0^{\sqrt{2}} \|m'(t)\| \, dt =$$

$$= \int_{\pi/4}^{\pi/2} \|(2\sin t, 2\cos t)\| \, dt + \int_0^2 \|(0, 1)\| \, dt + \int_0^{\sqrt{2}} \|(1, 1)\| \, dt =$$

$$= \int_{\pi/4}^{\pi/2} \sqrt{4\sin^2 t + 4\cos^2 t} \, dt + \int_0^2 1 \, dt + \int_0^{\sqrt{2}} \sqrt{2} \, dt =$$

$$= 2 \left(\frac{\pi}{2} - \frac{\pi}{4} \right) + 2 + \sqrt{2} \cdot \sqrt{2} = 2 \cdot \frac{\pi}{4} + 4 = \frac{\pi}{2} + 4$$

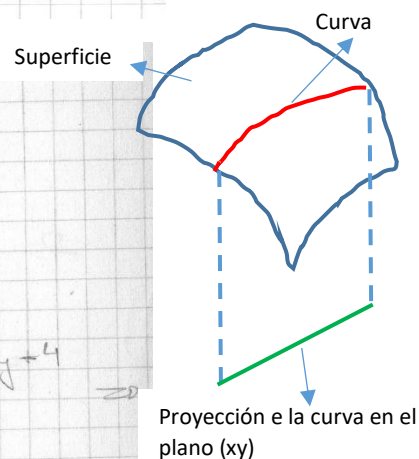
03) $z = x^2 - 4y^2$

por el (xy) $\rightarrow$ segmento $(1, 2, 0) \rightarrow (3, 1, 0)$

$$(x, y, z) = (1, 2, 0) + t[(3, 1, 0) - (1, 2, 0)]$$

$$(x, y, z) = (1, 2, 0) + t(2, -1, 0)$$

$$\begin{cases} x = 1+2t \\ y = 2-t \\ z = 0 \end{cases} \Rightarrow \begin{cases} \frac{x-1}{2} = \frac{y-2}{-1} \\ z = 0 \end{cases} \Rightarrow \begin{cases} -x+1 = 2y+4 \\ z = 0 \end{cases}$$



$$\Rightarrow \begin{cases} x = -2y + 4 + 1 \\ z = 0 \end{cases} \Rightarrow \begin{cases} x = -2y + 5 \\ z = 0 \end{cases} \quad (*)$$

Reempl con (*) en $z = x^2 - 4y^2$

$$z = (-2y + 5)^2 - 4y^2 = \cancel{4y^2} + 25 - 20y - \cancel{4y^2} = 25 - 20y$$

$$\vec{g}(y) = (-2y + 5, y, 25 - 20y) \quad 2 \geq y \geq 1$$

$$\vec{g}(y_0) = (1, 2, 19) \Rightarrow y_0 = 2 \quad (\text{satisface todas las ecuaciones})$$

$$\vec{g}(y_0) = (3, 1, 5) \Rightarrow y_0 = 1 \quad (\text{satisface todas las ecuaciones})$$

$$L = \int_1^2 \|\vec{g}'(y)\| \, dy = \int_1^2 \|(-2, 1, -20)\| \, dy =$$

$$= \int_1^2 \sqrt{4 + 1 + 400} \, dy = \sqrt{405}$$

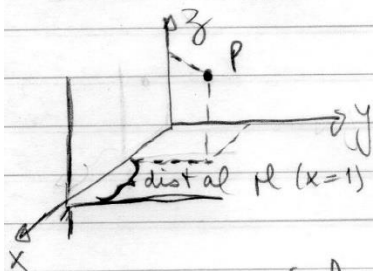
$$4) \begin{cases} y=x \\ 2y=x^2+z \rightarrow z=2y-x^2=2x-x^2 \\ z \geq 0 \end{cases}$$

$$\bar{g}(t) = (t, t, 2t - t^2) \quad \bar{g}(t_0) = (0, 0, 0) \Rightarrow t_0 = 0$$

$$\bar{g}(t_1) = (1, 1, 1) \Rightarrow t_1 = 1$$

Fluente $\rightarrow \bar{g}: [0, 1] \rightarrow \mathbb{R}^3 / \bar{g}(t) = (t, t, 2t - t^2)$

Resistencia



$$\text{dist} = |x-1| \Rightarrow f(x, y, z) = k|x-1|$$

$$\Omega = \int_C f(x, y, z) \, ds = \int_C k|x-1| \, ds =$$

$$= \int_0^1 \underbrace{k|t-1|}_{f(\bar{g}(t))} \underbrace{\|\bar{g}'(t)\|}_{ds} \, dt = \int_0^1 \underbrace{k(1-t)}_{\text{porque } 0 \leq t \leq 1, \text{ luego } |t-1| = 1-t} \cdot \|\bar{g}'(t)\| \, dt =$$

$$= \int_0^1 k(1-t) \|(1, 1, 2-2t)\| \, dt = \int_0^1 k(1-t) \sqrt{1+1+(2-2t)^2} \, dt =$$

$$= \int_0^1 k(1-t) \sqrt{2+4+4t^2-8t} \, dt = k \int_0^1 (1-t) \sqrt{4t^2-8t+6} \, dt =$$

$$= k \left(\frac{\sqrt{3}}{\sqrt{2}} - \frac{\sqrt{2}}{6} \right)$$

5) 1^{st} attempt

$$\begin{cases} x+y+z=4 \\ y=2x \end{cases} \Rightarrow \begin{cases} z=4-x-y \\ y=2x \end{cases} \Rightarrow \begin{cases} z=4-x-2x \\ y=2x \end{cases}$$

$$\vec{g}(x) = (x, 2x, 4-3x) \Rightarrow \vec{g}'(x) = (1, 2, -3)$$

$$x \geq 0 \wedge 2x \geq 0 \wedge 4-3x \geq 0$$

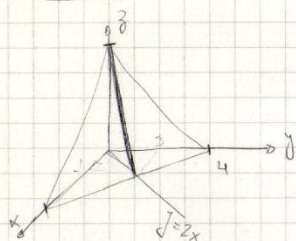
$$\downarrow$$

$$x \geq 0$$

$$\downarrow$$

$$4 \geq 3x \Rightarrow x \leq \frac{4}{3}$$

$$0 \leq x \leq \frac{4}{3}$$



dist. to a eye $z = \sqrt{x^2 + y^2}$
 $s = R \sqrt{x^2 + y^2} \quad (R \in \mathbb{R}^+)$

$$x_G = \frac{M_{xG}}{m_{\text{area}}}$$

$$y_G = \frac{M_{yG}}{m}$$

$$z_G = \frac{M_{zG}}{m}$$

$$x_G = \frac{\int_0^{4/3} \int_0^{4-3x} x \|\vec{g}'(x)\| dy dx}{\int_0^{4/3} \int_0^{4-3x} \|\vec{g}'(x)\| dy dx} = \frac{\int_0^{4/3} x \sqrt{1+4+9} \int_0^{4-3x} k \sqrt{x^2+4x^2} dx}{\int_0^{4/3} \sqrt{14} \int_0^{4-3x} k \sqrt{x^2+4x^2} dx} =$$

$$= \frac{k \sqrt{14} \int_0^{4/3} x \sqrt{5x^2} dx}{k \sqrt{14} \int_0^{4/3} \sqrt{5} \int_0^{4-3x} |x| dx} = \frac{\sqrt{5} \int_0^{4/3} x^2 dx}{\sqrt{5} \int_0^{4/3} x dx} =$$

$$= \frac{\left[\frac{x^3}{3} \right]_0^{4/3}}{\left[\frac{x^2}{2} \right]_0^{4/3}} = \frac{\frac{1}{3} \cdot \frac{4^3}{27}}{\frac{1}{2} \cdot \frac{4^2}{9}} = \frac{4}{9} \cdot 2 = \frac{8}{9}$$

$$y_G = \frac{\int_0^{4/3} \int_0^{4-3x} y \|\vec{g}'(x)\| dy dx}{\int_0^{4/3} \int_0^{4-3x} \|\vec{g}'(x)\| dy dx} = \frac{\int_0^{4/3} \sqrt{14} \cdot x \sqrt{5x^2} \cdot 2x dx}{k \sqrt{14} \sqrt{5} \cdot \frac{4^2}{18}} =$$

$$= \frac{2 \int_0^{4/3} x^2 dx}{\frac{8^2}{9}} = \frac{2 \left[\frac{x^3}{3} \right]_0^{4/3}}{\frac{64}{9}} = \frac{2}{4} \cdot \frac{4^3}{27} \cdot \frac{1}{3} = \frac{16}{9}$$

$$\begin{aligned}
 \beta_6 &= \frac{\int_0^{4/3} \int \|\vec{g}'(x)\| dx}{\int_0^{4/3} \|\vec{g}'(x)\| dx} = \frac{\int_0^{4/3} k \sqrt{5x^2} (4-3x) \sqrt{4} dx}{k \sqrt{4} \sqrt{5} \cdot \frac{16}{18}} = \\
 &= \frac{9}{8} \int_0^{4/3} (4x - 3x^2) dx = \frac{9}{8} \left[2x^2 - x^3 \right]_0^{4/3} = \\
 &= \frac{9}{8} \left[2 \cdot \frac{16}{9} - \frac{\overbrace{4^3}^{64}}{27} \right] = \frac{1}{1} \left[\frac{32}{3} - \frac{64}{27} \right] = \frac{4}{3} \\
 G &= \left(\frac{8}{9}, \frac{16}{9}, \frac{4}{3} \right)
 \end{aligned}$$

06) $\vec{g}$ derivable, $\|\vec{g}\| = k$ ($k \in \mathbb{R}$)

Efectuamos la derivada del sig. producto escalar:

$$\left. \begin{aligned}
 \text{i)} (\vec{g} \cdot \vec{g})' &= \vec{g}' \cdot \vec{g} + \vec{g} \cdot \vec{g}' = 2 \vec{g} \cdot \vec{g}' \\
 \text{ii)} (\vec{g} \cdot \vec{g})' &= (\|\vec{g}\| \|\vec{g}\| \cos \alpha)' = (k^2 \cos \alpha)' = 0
 \end{aligned} \right\} \Rightarrow$$

$$\Rightarrow 2 \vec{g} \cdot \vec{g}' = 0 \Rightarrow \vec{g} \cdot \vec{g}' = 0 \Rightarrow \vec{g} \perp \vec{g}'$$

$$8) \int_C \vec{F} \cdot d\vec{s} = \int_C (P, Q) \cdot d\vec{s} = \textcircled{1}$$

$$\vec{s} = (x(t), y(t))$$

$$d\vec{s} = (x'(t), y'(t)) dt = (x' dt, y' dt) = (dx, dy)$$

$$\text{ya que si } \begin{cases} x = x(t) \\ y = y(t) \end{cases} \Rightarrow \begin{cases} dx = x' dt \\ dy = y' dt \end{cases}$$

$$\textcircled{1} = \int_C (P, Q) \cdot (dx, dy) = \int_C P dx + Q dy$$

$$\text{En } \mathbb{R}^3: \vec{F} = (P, Q, R)$$

$$\vec{s} = (x(t), y(t), z(t))$$

$$d\vec{s} = (x'(t), y'(t), z'(t)) dt = (x' dt, y' dt, z' dt) = (dx, dy, dz)$$

$$\text{ya que si } \begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases} \Rightarrow \begin{cases} dx = x' dt \\ dy = y' dt \\ dz = z' dt \end{cases}$$

$$\int_C \vec{F} \cdot d\vec{s} = \int_C (P, Q, R) \cdot (dx, dy, dz) = \int_C P dx + Q dy + R dz$$

$$9) \mathcal{L} = \int_C \vec{F} \cdot d\vec{s} \quad \vec{F}(\vec{r}) = \vec{k} \quad (\vec{k} \text{ un vector constante})$$

$$\text{segmento } \overrightarrow{AB} \\ \vec{g}: [0, 1] \rightarrow \mathbb{R}^2 / \vec{g}(t) = \vec{A} + t(\vec{B} - \vec{A})$$

$$\mathcal{L}_{AB} = \int_C \vec{F} \cdot d\vec{s} = \int_C \vec{F} \cdot d\vec{g} = \int_C \vec{k} \cdot \vec{g}'(t) dt =$$

$$= \int_0^1 \vec{k} \cdot (\vec{B} - \vec{A}) dt = \vec{k} (\vec{B} - \vec{A}) \int_0^1 dt = \vec{k} (\vec{B} - \vec{A}) = \underbrace{\vec{F}}_{\text{fuerza}} \underbrace{(\vec{B} - \vec{A})}_{\text{distancia}} = \text{fuerza} \times \text{dist}$$

$$\begin{aligned}
 10) \quad \int_C \bar{H} \, d\bar{s} &= \int_C k \bar{T} \, d\bar{s} = k \int_C \bar{T} \, d\bar{s} = k \int_C \frac{\bar{q}'(t)}{\|\bar{q}'(t)\|} \cdot \bar{q}'(t) \, dt = \\
 &= k \int_C \frac{\|\bar{q}'(t)\|^2}{\|\bar{q}'(t)\|} \, dt = k \int_C \|\bar{q}'(t)\| \, dt = k \int_a^b \|\bar{q}'(t)\| \, dt
 \end{aligned}$$

Sei $\bar{q}: [a, b] \rightarrow \mathbb{R}^n$ la parametrización de C

$$\text{Si } \bar{H} = k \bar{T} \Rightarrow \|\bar{H}\| = \|k \bar{T}\| = |k| \|\bar{T}\| = k \cdot 1 = k$$

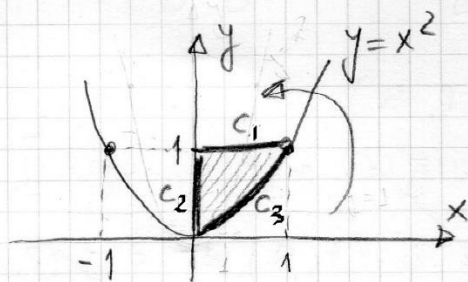
$$\text{Luego } \|\bar{H}\| = k$$

$$\text{Por lo tanto } k \int_a^b \|\bar{q}'(t)\| \, dt = \|\bar{H}\| \text{ long } C$$

19)

$$x^2 \leq y \leq 1$$

$$0 \leq x \leq 1$$



$$C = C_1 \cup C_2 \cup C_3$$

$$C_1: \vec{r}_1(t) = (t, 1) \quad 1 \geq t \geq 0$$

$$C_2: \vec{r}_2(t) = (0, t) \quad 1 \geq t \geq 0$$

$$C_3: \vec{r}_3(t) = (t, t^2) \quad 0 \leq t \leq 1$$

$$\vec{f}(x, y) = (y, -x)$$

$$\int_C \vec{f} \cdot d\vec{s} = \int_{C_1} \vec{f} \cdot d\vec{s} + \int_{C_2} \vec{f} \cdot d\vec{s} + \int_{C_3} \vec{f} \cdot d\vec{s} = \left[\int_1^0 (1, -t) \cdot (1, 0) dt \right] + \left[\int_1^0 (t, 0) \cdot (0, 1) dt \right] +$$

$$\left[\int_0^1 (t^2, -t) \cdot (1, 2t) dt \right] = \int_1^0 1 dt + \int_1^0 0 dt + \int_0^1 (t^2 - 2t^2) dt =$$

$$= -1 + \left[-\frac{t^3}{3} \Big|_0^1 \right] = -1 + \left(-\frac{1}{3} \right) = -\frac{4}{3}$$

12) Circulación: $\int_C \vec{f} \cdot d\vec{s}$ con $\vec{f}(\vec{x}) = (x-y, x, yz)$

$$C: \begin{cases} z = x - y^2 \\ y = x^2 \end{cases} \Rightarrow \begin{cases} z = x - x^4 \\ y = x^2 \end{cases}$$

$$\vec{g}(t) = (t, t^2, t - t^4)$$

$$\begin{array}{l} \text{Punto inicial: } (1, 1, 0) \quad \vec{g}(t_0) = (1, 1, 0) \Rightarrow t_0 = 1 \\ \text{Punto final: } (-1, 1, -2) \quad \vec{g}(t_1) = (-1, 1, -2) \Rightarrow t_1 = -1 \end{array} \quad 1 \geq t \geq -1$$

$$\int_C \vec{f} \cdot d\vec{s} = \int_1^{-1} \vec{f}(\vec{g}(t)) \cdot \vec{g}'(t) dt =$$

$$= \int_1^{-1} (t - t^2, t, t^2(t - t^4)) \cdot (1, 2t, 1 - 4t^3) dt =$$

$$= \int_1^{-1} (t - t^2, t, t^3 - t^6) \cdot (1, 2t, 1 - 4t^3) dt =$$

$$= \int_1^{-1} [t - t^2 + 2t^2 + (t^3 - t^6)(1 - 4t^3)] dt =$$

$$= \int_1^{-1} (t + t^2 + t^3 - 5t^6 + 4t^9) dt =$$

$$= \left[\frac{t^2}{2} + \frac{t^3}{3} + \frac{t^4}{4} - \frac{5t^7}{7} + \frac{4t^{10}}{10} \right]_1^{-1} = \left(\frac{16}{21} \right)$$

13) pto inicial $\rightarrow \bar{X}(1) = (0, 1, 2)$

pto final $\rightarrow \bar{X}(3) = (2, 9, 6)$

$$\sum_{AB} = \int_1^3 \vec{f} \cdot d\vec{s} = \int_1^3 [3(t-1), -(t-1)2t, t^2 2t] \cdot$$

$$(1, 2t, 2) dt = \int_1^3 [3t-3-2t(t-1)2t+4t^3] dt =$$

$$= \int_1^3 (3t-3-4t^2+4t^3) dt = \left[\frac{4t^3}{3} + \frac{3t^2}{2} - 3t \right]_1^3 =$$

$$= \frac{4 \cdot 27}{3} + \frac{3 \cdot 9}{2} - 9 - \frac{4}{3} - \frac{3}{2} + 3 = \frac{122}{3}$$

Para asegurar la independencia del camino, verifiquemos si existe p. potencial:

$$P(x, y, z) = 3x$$

$$Q(x, y, z) = -xz$$

$$R(x, y, z) = yz$$

$$\vec{f}(x, y, z) = (3x, -xz, yz)$$

$\frac{d}{dx} \phi_x?$ $\frac{d}{dy} \phi_y?$ $\frac{d}{dz} \phi_z?$

$$\left. \begin{matrix} P'_y = 0 \\ Q'_x = -z \end{matrix} \right\} \neq$$

$$\left. \begin{matrix} P'_z = 0 \\ R'_x = 0 \end{matrix} \right\}$$

$$\left. \begin{matrix} Q'_z = -x \\ R'_y = z \end{matrix} \right\} \neq$$

No existe p. potencial, por lo tanto el campo no es conservativo (el dominio $\mathbb{R}^3$ es simplemente conexo).

Si $\vec{f}$ no es conservativo, el resultado de la integral no es independiente del camino.

14)

a) $P(x, y) = y - 2xy + 1$

$Q(x, y) = x + 1 - x^2$

$$\left. \begin{array}{l} P'_y = 1 - 2x \\ Q'_x = 1 - 2x \end{array} \right\} \Rightarrow \text{Dom } \bar{f} = \mathbb{R}^2 \text{ simplesmente conexo}$$

$$\phi = \int (y - 2xy + 1) dx = yx - x^2y + x + g(y)$$

$$\left. \begin{array}{l} \phi'_y = x - x^2 + g'(y) \\ \phi'_x = x + 1 - x^2 \end{array} \right\} \Rightarrow g'(y) = 1 \Rightarrow \frac{dg}{dy} = 1 \Rightarrow \int dg = \int dy \Rightarrow$$

$$\Rightarrow g = y + c$$

$$\boxed{\phi = yx - x^2y + x + y + c}$$

b) $P(x, y) = x - y^2$

$Q(x, y) = y - x^2$

$$\left. \begin{array}{l} P'_y = -2y \\ Q'_x = -2x \end{array} \right\} \neq \text{no existe f. potencial}$$

14) c) $P = z \cos(xz)$

$Q = z$

$R = y + x \cos(xz)$

$$P'_y = 0 \quad Q'_x = 0$$

condic. necess.
 satisfeitas

$$P'_z = \cos(xz) - xz \sin(xz) \quad R'_x = \cos(xz) - xz \sin(xz)$$

$$Q'_z = 1 \quad R'_y = 1$$

Dom $\bar{f} = \mathbb{R}^3$ simplesmente conexo

$$\phi = \int Q dy = \int z dy = zy + h(x, z)$$

$$\left. \begin{array}{l} \phi'_x = h'_x \\ \phi'_x = z \cos(xz) \end{array} \right\} \Rightarrow h'_x = z \cos(xz)$$

$$\Rightarrow h(x, z) = \int z \cos(xz) dx = \frac{z \sin(xz)}{z} + g(z)$$

$$\Rightarrow \phi = zy + \sin(xz) + g(z)$$

$$\left. \begin{array}{l} \phi'_z = y + \cos(xz) \cdot x + g'(z) \\ \phi'_z = y + x \cos(xz) \end{array} \right\} \Rightarrow g'(z) = 0 \Rightarrow g(z) = k$$

$$\phi'_z = y + x \cos(xz)$$

$$\boxed{\phi = zy + \sin(xz) + k}$$

$$d) \quad P = 2x + y + 1$$

$$Q = x + z$$

$$R = y + 2z$$

$$\vec{f}(x, y, z) = (\underbrace{2x+y+1}_P, \underbrace{x+z}_Q, \underbrace{y+2z}_R)$$

$$P'_x = 1$$

$$P'_z = 0$$

$$Q'_z = 1$$

$$Q'_x = 1$$

$$R'_x = 0$$

$$R'_y = 1$$

} amplo condic
necesária

Dom $\vec{f} = \mathbb{R}^3$ simplesmente conexo $\rightarrow$ amplo condic suficiente

$$\phi = \int (2x + y + 1) dx = x^2 + xy + x + g(y, z)$$

$$\left. \begin{aligned} \phi'_y &= x + g'_y \\ \phi'_z &= x + z \end{aligned} \right\} \Rightarrow g'_y = z \Rightarrow \int dg_y = \int z dy \Rightarrow$$

$$\Rightarrow g(y, z) = zy + h(z)$$

$$\phi = x^2 + xy + x + zy + h(z)$$

$$\left. \begin{aligned} \phi'_z &= y + h'(z) \\ \phi'_z &= y + 2z \end{aligned} \right\} \Rightarrow h'(z) = 2z \Rightarrow \int dh = \int 2z dz \Rightarrow$$

$$\Rightarrow h(z) = z^2 + c$$

$$\boxed{\phi = x^2 + xy + x + zy + z^2 + c}$$

$$15) \quad \vec{f} = \nabla \phi ; \quad C_{AB} \rightarrow \int_C \vec{f} \cdot d\vec{s} = \phi(B) - \phi(A)$$

$$\vec{f} = \nabla \phi \wedge U = -\phi$$

$$\vec{f} = \nabla \phi \wedge -U = \phi \Rightarrow \nabla \phi = \nabla(-U) = -\nabla U \Rightarrow \\ \Rightarrow \vec{f} = -\nabla U$$

$$-\int_C \vec{f} \cdot d\vec{s} = -[\phi(B) - \phi(A)] = -\phi(B) + \phi(A) = \\ = \phi(A) - \phi(B)$$

$$16) \quad D\vec{f} = \begin{bmatrix} \frac{-2xy}{(x^2+y^2)^2} & \frac{x^2-y^2-2y^2}{(x^2+y^2)^2} \\ \frac{-x^2-y^2+2x^2}{(x^2+y^2)^2} & \frac{+2xy}{(x^2+y^2)^2} \end{bmatrix}$$

$$\vec{f} = \frac{1}{x^2+y^2} \begin{pmatrix} -2xy & x^2-y^2 \\ x^2-y^2 & 2xy \end{pmatrix}$$

$\vec{f}$ es continua en $\mathbb{R}^2 - \{0\}$ ya sea cociente de funciones continuas (polinómicas) donde el denominador no se anula nunca.

$$(\vec{f})^t = \vec{f}, \text{ por lo tanto } \vec{f} \text{ es simétrica}$$

Sea $C: x^2+y^2=1$, descrita por:

$$\vec{q}: [0, 2\pi] \rightarrow \mathbb{R}^2 / \vec{q}(t) = (\cos t, \sin t)$$

$$\text{Calculamos } \oint_C \vec{f} \cdot d\vec{s} = \int_0^{2\pi} \vec{f}(\vec{q}(t)) \cdot d\vec{s} =$$

$$= \int_0^{2\pi} \left(\frac{\sin t}{\cos^2 t + \sin^2 t}, \frac{-\cos t}{\sin^2 t + \cos^2 t} \right) \cdot (-\sin t, \cos t) dt =$$

$$= \int_0^{2\pi} (\sin^2 t - \cos^2 t) dt = - \int_0^{2\pi} dt = -2\pi \neq 0$$

Si $\vec{f}$ admitiera p. potencial; la integral de $\vec{f}$ a lo largo de cualquier curva cerrada debería dar 0. Sin embargo, en este caso $\oint_C \vec{f} \cdot d\vec{s} = -2\pi \neq 0$, y por lo tanto $\vec{f}$ no admite p. potencial.

$$17) \quad \bar{f} = (f_1, f_2) = (P, Q)$$

$$\left. \begin{array}{l} P'_x = 2y - 6x \\ Q'_x = 2x \end{array} \right\} \begin{array}{l} P'_y = 2x \\ Q'_y = 0 \end{array} \quad \left. \begin{array}{l} \text{condic} \\ \text{necesaria} \end{array} \right\} \text{Dom } \bar{f} = \mathbb{R}^2 \text{ simpl convexo}$$

$$P = \int (2y - 6x) dx = 2xy - 3x^2 + h(y)$$

$$\left. \begin{array}{l} P'_y = 2x + h'_y \\ P'_y = 2x \end{array} \right\} \Rightarrow h'_y = 0 \Rightarrow h(y) = c$$

$$P = 2xy - 3x^2 + c$$

$$Q = \int 2x dx = x^2 + m(y)$$

$$\left. \begin{array}{l} Q'_y = m'_y \\ Q'_y = 0 \end{array} \right\} \Rightarrow m'_y = 0 \Rightarrow m(y) = k$$

$$Q = x^2 + k$$

$$\bar{f} = (2xy - 3x^2 + c, x^2 + k)$$

$$\Phi = \int (2xy - 3x^2 + c) dx = x^2y - x^3 + cx + t(y)$$

$$\left. \begin{array}{l} \Phi'_y = x^2 + t'_y \\ \Phi'_y = x^2 + k \end{array} \right\} \Rightarrow t'_y = k \Rightarrow t(y) = ky + a$$

$$\boxed{\Phi = x^2y - x^3 + cx + ky + a}$$

(A)

$$\phi(1,1) = 3 \Rightarrow 3 = 1 - 1 + c + k + a \Rightarrow 3 = c + k + a \quad (*)$$

$$\phi'_x = 2xy - 3x^2 + c \quad \phi'_x(1,1) = 0 \Rightarrow 2 - 3 + c = 0 \Rightarrow c = 1$$

$$\phi'_y = x^2 + k \quad \phi'_y(1,1) = 1 \Rightarrow 1 + k = 1 \Rightarrow k = 0$$

Reemplazando en $(*) \rightarrow 1 + 0 + a = 3 \Rightarrow a = 2$

Reemplazando en $(A) \rightarrow \boxed{\phi = x^2 y - x^3 + x + 2}$

A pertenece al eje y $\rightarrow A = (0, a) \quad a \in \mathbb{R}$

B pertenece a la curva $y = x - x^{-1} \rightarrow B = (b, b - b^{-1}) \quad b \in \mathbb{R}$

$$\int_C \vec{f} \cdot d\vec{g} = \int_A^B \vec{f} \cdot d\vec{g} = \phi(B) - \phi(A) = b^2(b - b^{-1}) - b^3 + b + 2 - 0 + 0 - 0 - 2 = \cancel{b^3} - \cancel{b} - \cancel{b^3} + \cancel{b} + \cancel{2} - \cancel{2} = 0$$

Sea $\vec{f}$ de clase C^1 / $\vec{f}(x,y) = (x y^2, y g(x))$. Determine g de manera que $\vec{f}$ admita función potencial. Suponga $\vec{f}(2,1) = (2,6)$

$$\begin{aligned} 18) \quad P &= x y^2 & P'_y &= 2xy \\ Q &= y g(x) & Q'_x &= y g'(x) \end{aligned} \quad \left. \vphantom{\begin{aligned} P &= x y^2 \\ Q &= y g(x) \end{aligned}} \right\} \Rightarrow$$

$$\Rightarrow 2xy = y g' \Rightarrow 2x = \frac{dg}{dx} = \int 2x dx = \int dg \Rightarrow$$

$$\Rightarrow g(x) = x^2 + C$$

$$\vec{f}(x,y) = [x y^2, y(x^2 + C)]$$

$$\vec{f}(2,1) = (2,6) \Rightarrow (2 \cdot 1^2, 1(4+C)) = (2,6) \Rightarrow$$

$$\Rightarrow 4+C=6 \Rightarrow C=2$$

Luego: $g(x) = x^2 + 2$

$$19) \quad \vec{f}(x,y,z) = (2xy + z^2, x^2, 2xz)$$

$$\frac{\partial(2xy + z^2)}{\partial y} = \frac{\partial(x^2)}{\partial x}$$

$$2x = 2x \quad (1)$$

$$\frac{\partial(2xy + z^2)}{\partial z} = \frac{\partial(2xz)}{\partial x}$$

$$2z = 2z \quad (2)$$

$$\frac{\partial(x^2)}{\partial z} = \frac{\partial(2xz)}{\partial y}$$

$$0 = 0 \quad (3)$$

Por (1), (2), (3)
condic. necesaria

Dom $\vec{f} = \mathbb{R}^3$
simplemente
conexo: condic. suficiente

$$\begin{cases} \phi'_x = 2xy + z^2 \\ \phi'_y = x^2 \\ \phi'_z = 2xz \end{cases}$$

$$\phi = \int x^2 dy = x^2 y + g(x, z) \Rightarrow$$

$$\Rightarrow \phi'_x = 2xy + g'_x = 2xy + z^2 \Rightarrow g'_x = z^2 \Rightarrow$$

$$\Rightarrow g(x, z) = z^2 x + h(z)$$

$$\phi(x, y, z) = x^2 y + z^2 x + h(z) \Rightarrow \phi'_z = 2xz + h'(z) = 2xz =$$

$$\Rightarrow h'(z) = 0 \Rightarrow h(z) = C$$

$$\phi(x, y, z) = x^2 y + z^2 x + C$$

$$\vec{x}(0) = (1, 0, 0)$$

$$\vec{x}(1) = (2+1, 0, 3) = (3, 0, 3)$$

$$\int_C \vec{f} \cdot d\vec{s} = \phi(3, 0, 3) - \phi(1, 0, 0) = [3^2 \cdot 0 + 3^2 \cdot 3 + C] - [1^2 \cdot 0 + 0^2 \cdot 1 + C] = (27)$$

$$20) \quad \vec{f} = k \vec{x}'(t) = k \vec{g}'(t) \quad (k \neq 0 \text{ para que sea proporcional})$$

$$\text{Sea } C: \vec{g}: [a, b] \rightarrow \mathbb{R}^2 / \vec{g} = \vec{g}(t)$$

$$\int_C \vec{f} \, ds = \int_a^b \vec{f} \cdot \vec{g}'(t) \, dt = \int_a^b k \vec{g}'(t) \cdot \vec{g}'(t) \, dt =$$

$$= k \int_a^b \|\vec{g}'(t)\| \, dt = k \text{ Long}(C)$$

Si C es una curva cerrada de long ℓ ($\ell \in \mathbb{R}$)

$$\text{resulta} \quad \int_C \vec{f} \, ds = k \cdot \ell \neq 0 \quad (k \neq 0 \wedge \ell \neq 0)$$

por long. de curva

$$21) \quad \vec{f}(x, y) = (ax, y - ax) \quad \vec{x}(t) = (\cos t, b \sin t)$$

$$\vec{x}'(t) = (-\sin t, b \cos t)$$

$$\int_C \vec{f} \, ds = \int_0^{2\pi} (a \cos t, b \sin t - a \cos t) \cdot (-\sin t, b \cos t) \, dt =$$

$$\int_0^{2\pi} \vec{f}(\vec{x}(t)) \cdot \vec{x}'(t) \, dt$$

$$= \int_0^{2\pi} (-a \cos t \sin t + b^2 \sin t \cos t - ab \cos^2 t) \, dt =$$

$$= \left[-a \frac{\sin^2 t}{2} \right]_0^{2\pi} + b^2 \left[\frac{\sin^2 t}{2} \right]_0^{2\pi} - ab \left[\frac{t}{2} + \right.$$

$$\left. + \frac{\sin 2t}{4} \right]_0^{2\pi} = 0 + 0 - ab \left[\frac{2\pi}{2} + 0 - 0 \right] =$$

$$= -ab\pi$$

$$g(a, b) = -ab\pi$$

$$a + b = 6 \Rightarrow b = 6 - a$$

$$h(a) = -a(6 - a)\pi = (-6a + a^2)\pi$$

$$h' = (-6 + 2a)\pi = 0 \Rightarrow -6 + 2a = 0 \Rightarrow a = \frac{6}{2} = 3$$

$$h'' = 2 > 0 \Rightarrow h(3) \text{ es m\u00ednimo}$$

$$\boxed{a = 3 \Rightarrow b = 6 - 3 = 3}$$